

ITHACA TOMPKINS, NY, USA

WMO: 725155

Lat: 42.483N

Lon: 76.467W

Elev: 1099

StdP: 14.12

Time Zone: -5.00 (NAE)

Period: 05-19

WBAN: 94761

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-0.5	4.6	-7.8	3.8	1.4	-3.1	4.8	6.1	27.1	20.8	25.1	23.3	7.9	80	0.538

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	88.3	72.4	85.5	71.0	82.7	69.5	74.9	84.5	73.2	82.0	71.6	79.4	8.8	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.8	122.4	79.6	70.4	116.4	78.2	69.0	111.0	76.7	38.9	84.4	37.5	81.8	36.1	79.5	81.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	23.5	20.1	18.0	DB	-8.9	93.7	5.1	2.5	-12.6	95.5	-15.5	96.9	-18.4	98.3	-22.1	100.2
(5)				WB	-9.1	77.8	5.0	1.8	-12.7	79.1	-15.6	80.1	-18.4	81.2	-22.1	82.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.4	23.9	24.6	32.4	45.2	57.5	65.1	70.0	68.1	61.6	50.2	38.9	30.0
	DBStd	18.73	11.61	11.09	11.33	10.14	9.07	6.82	5.52	5.58	7.59	8.83	9.31	9.07
	HDD50	3402	810	712	559	208	30	0	0	0	3	109	350	621
	HDD65	6898	1273	1130	1011	598	267	82	18	33	154	464	784	1084
	CDD50	2465	2	2	13	65	263	454	619	561	352	116	16	2
	CDD65	488	0	0	0	5	36	86	172	130	53	6	0	0
	CDH74	4267	0	0	2	70	391	781	1525	1017	444	37	0	0
	CDH80	1218	0	0	0	19	96	226	482	269	123	3	0	0
Wind	WSAvg	7.8	9.6	9.8	9.3	9.1	7.1	6.2	5.5	5.5	5.9	7.3	8.6	9.3
Precipitation	PrecAvg	36.1	2.0	1.9	2.4	3.1	3.3	3.8	3.5	3.4	3.8	3.1	3.1	2.4
	PrecMax	47.8	6.4	3.8	5.5	8.1	8.1	11.8	8.4	8.5	9.1	8.3	6.4	4.9
	PrecMin	25.4	0.5	0.2	0.4	0.8	0.9	1.1	0.9	1.1	0.7	0.2	0.7	0.4
	PrecStd	5.8	1.1	0.9	1.0	1.5	1.5	1.7	1.8	1.5	1.7	1.8	1.3	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.8	62.1	71.6	82.3	87.3	90.3	91.9	90.3	88.9	78.8	67.5	60.5
		MCWB	52.8	53.5	56.6	61.1	69.1	73.5	74.5	73.7	72.0	67.5	56.8	56.4
	2%	DB	51.7	51.9	61.5	74.2	83.1	86.2	88.3	86.4	83.5	72.9	61.7	52.4
		MCWB	46.1	44.8	50.7	58.4	67.3	71.5	72.8	71.8	71.0	63.1	53.0	48.5
	5%	DB	45.8	46.2	55.6	67.8	78.7	82.4	85.3	82.7	79.4	68.8	57.6	47.1
		MCWB	42.0	41.7	48.4	54.4	65.2	69.5	71.3	69.9	68.5	62.0	50.1	43.8
	10%	DB	40.1	40.0	50.0	62.5	74.2	78.8	82.3	80.0	74.9	64.3	53.6	43.2
		MCWB	37.2	35.8	43.5	52.3	62.2	66.9	70.0	68.6	66.2	58.6	48.0	40.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	55.4	58.8	64.3	72.7	75.3	77.2	76.4	75.1	69.7	58.4	56.1
		MCDB	57.6	60.9	67.7	77.0	82.3	86.6	87.7	86.0	84.5	75.3	63.6	59.9
	2%	WB	47.5	46.2	53.4	60.1	69.7	73.2	75.1	74.1	72.4	65.9	55.1	49.0
		MCDB	49.6	50.8	59.9	70.5	80.9	83.7	85.1	82.2	80.8	70.2	60.4	51.8
	5%	WB	41.9	41.7	48.9	56.8	66.9	71.1	73.3	72.2	70.0	62.9	51.6	44.4
		MCDB	45.1	46.3	55.0	65.8	75.3	79.6	82.1	79.6	77.4	67.2	56.1	47.0
	10%	WB	37.0	36.1	43.7	53.3	64.3	69.0	71.6	70.4	67.5	59.3	48.3	40.5
		MCDB	39.3	39.9	49.6	61.1	71.8	76.1	78.9	77.2	72.6	64.1	53.0	43.0
Mean Daily Temperature Range	5% DB	MDBR	14.7	15.5	17.7	20.7	21.8	20.4	21.3	20.5	20.5	17.3	15.5	12.4
		MCDBR	18.0	21.9	24.8	29.9	26.5	25.2	24.5	23.5	24.2	22.6	22.1	17.3
		MCWBR	16.0	17.4	16.5	16.7	13.5	12.8	11.1	10.9	12.1	13.8	15.0	15.1
	5% WB	MCDBR	17.4	21.1	22.2	25.3	22.9	21.4	20.6	20.2	21.4	19.0	18.3	17.1
	MCWBR	16.2	17.8	16.2	16.2	12.7	11.9	10.4	10.4	11.5	13.2	14.4	15.6	
Clear-Sky Solar Irradiance	taub	0.295	0.312	0.334	0.361	0.385	0.435	0.441	0.426	0.364	0.349	0.307	0.287	
	taud	2.372	2.325	2.378	2.364	2.335	2.214	2.210	2.240	2.398	2.442	2.514	2.468	
	Ebn at Noon	268	281	289	287	282	267	264	264	275	265	264	263	
	Edh at Noon	25	31	34	37	39	44	44	42	33	28	22	21	
All-Sky Solar Radiation	RadAvg	440	715	1050	1383	1667	1767	1809	1613	1289	804	519	351	
	RadStd	40	103	95	162	164	164	141	93	129	88	66	49	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+169	N/A

Nomenclature: See separate page